Abstract

Working memory temporarily stores and manipulates information, often modeled by recurrent dynamic networks that settle into stable states called attractors. These attractors represent memories by maintaining activity patterns even after stimulus removal. Short-term synaptic plasticity (STSP), including facilitation (STF) and depression (STD), modulates synaptic strength based on recent activity, supporting both memory stability and flexibility. Traditional attractor models like Hopfield networks lack synaptic plasticity, so we extend this framework by integrating STSP and modulation to better capture dynamic memory processes. Our research question is to explore how the balance between short-term facilitation and depression properties of synaptic plasticity shapes the functional trade-off between stability and flexibility in working memory. We first anticipate that facilitation-dominant networks will promote persistent, stable attractor states, supporting maintenance of memory. Secondly, we anticipate that depression-dominant networks will promote faster updating, enabling flexibility, yet introduce destabilisation at the cost of memory longevity. Our model is a biologically inspired, rate-based recurrent neural network grounded in attractor network theory. It simulates a two-excitatory and one-inhibitory attractor network, incorporating STF and STD based on Dayan & Abbott's model. Hebbian plasticity governs synaptic weight updates, enabling activity-dependent connectivity that reinforces stable patterns (attractors). A memory is represented by such a stable pattern, and successful encoding is indicated when the network maintains or returns to this pattern upon stimulus presentation. In preliminary explorations, we defined three synaptic weights – excitation-to-excitation, excitation-to-inhibition, and inhibition-to-excitation – that shape the attractor landscape. Adjusting these weights alters memory dynamics: increasing excitation-to-excitation promotes multi-stability, while stronger excitation-to-inhibition suppresses fixed states, preventing memory formation. We also examined how transient stimuli affect excitatory activity during working memory. Additionally, incorporating plasticity (STF and STD) revealed a significant link between their strengths and the balance between memory stability and flexibility. Future work should validate the model against experimental working memory data to assess its biological relevance. Expanding to larger or layered recurrent networks could test whether the stability-flexibility trade-off persists in high-dimensional systems.

A. What is the phenomena? Here summarize the part of the phenomena which your modeling addresses.

Working memory is a cognitive function that enables the temporary storage and manipulation of information. Computationally, it can be modeled through recurrent dynamic networks, which settle into stable states, coined "attractors." Activity can settle into attractors, allowing the information to persist even after removal of information, corresponding to a memory representation.

Short-term plasticity (STP) refers to rapid, activity-dependent changes in synaptic strength. Aspects of STP include short-term facilitation (STF) and depression (STD). STF leads to *synaptic enhancement and boosts recurrent excitation when activity is high; whereas STD suppresses neurotransmitter release, gradually weakens excitation, and allows for easier transitions. STP is thought to play a role in short-term memory and adaptation to incoming stimuli.*

The value of attractor models as powerful tools for probing the neural basis of cognition has been highlighted, yet traditional conceptualisations, like Hopfield networks, do not integrate emergent synaptic plasticity. There has been a call for better understanding of the mechanisms that control working memory.*Therefore, we offer an extension of attractor network theory to include STP and modulation. STP could support both stability (memory maintenance) and flexibility (updating memory), capturing the dynamic balance – and tradeoff – between maintaining or switching memories*

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B. What is the key scientific question?: Clearly articulate the question which your modeling tries to answer.

Our research question is to explore how the balance between short-term facilitation and depression properties of synaptic plasticity shapes the functional trade-off between stability and flexibility in working memory.

C. What was our hypothesis?: Explain the key relationships which we relied on to simulate the phenomena.

Our hypotheses are two-fold. First we anticipate that facilitation-dominant networks will promote persistent, stable attractor states, supporting maintenance of memory. Secondly, we also anticipate that depression-dominant networks will promote faster updating, enabling flexibility, yet introduce destabilisation which will be at the cost of memory longevity.

D. How did your modeling work? Give an overview of the model, it's main components, and how the modeling works. ''Here we … ''

Our model is designed to be a biologically inspired rate-based recurrent neural network that is grounded in the attractor network theory, implementing attractor dynamics. The main composition of the model is a simulation of two excitatory neurons and one inhibitory neuron. We input short-term facilitation and depression into our model, following Dayan & Abbott's model of synaptic plasticity, simulating how synaptic efficacy changes over short timescales due to recent activity. Another component of the model is that we implement the Hebbian plasticity rule, which updates synaptic weights based on the activity of the neurons, which is applied to update the connectivity between them. This activity-dependent connectivity reflects ongoing synaptic plasticity allowing the network to adaptively reinforce stable patterns (attractors). In our model, a memory would consist of a stable pattern. Upon stimulus presentation, if the memory is successfully encoded, the network remains in or returns to the same activity pattern – this is the attractor.

E. What did you find? Did the modeling work? Explain the key outcomes of your modeling evaluation.
Our initial experiments consisted of simulating a working memory based mechanism via a

two-excitatory and one-inhibitory unit attractor network. We simulated this multiple times with varying initial conditions and varying network weights. The weights denote the strength of connections between excitation-to-excitation, excitation-to-inhibition and inhibition-to-excitation. We found that given the characteristics of these three connections, we can modify the attractor landscape . For example, the excitation-to-excitation weight being responsible for positive feedback leads to stronger self and cross-excitation as it is increased. From our simulations, we see that increasing this parameter led to multi-stability and both memory states (corresponding to the two excitatory units) remaining active simultaneously. Increasing excitation-to-inhibition connection weight, on the other hand, leads to higher inhibition towards excitations leading to no fixed states and hence no memory. Building on to these simulations, we tested how addition of a transient stimulus affects the excitatory activities over time. Additionally, we ran some experiments that show the activities change when stimulus is switched between the units. Incorporation of short term synaptic plasticity in the form of short term facilitation (STF) and short term depression (STD) in the attractor model was also tested with respect to their effect on the stability and flexibility of the model's memory. From our preliminary exploration, we saw that there exists a clear pattern in the strength of memory stability and flexibility with respect to changing STF and STD strengths.

F. What can you conclude? Conclude as much as you can *with reference to the hypothesis*, within the limits of the modeling.
We aim to explore how the balance between STF and STD shapes the functional trade-off between stability and flexibility in working memory. For that, we design a biologically inspired rate-based attractor model, following Dayan & Abbott's model of synaptic plasticity. In our model, a memory would consist of a stable pattern. Upon stimulus, the network activates a specific neuron, which becomes self-sustaining post-stimulus. Our model shows that short-term facilitation promotes memory flexibility by allowing faster updating between states, while short-term depression enhances memory stability by reinforcing persistent activity patterns. Future directions should look to validate this model on real experimental data for working memory to explore if the model accurately reflects a system akin to working memory dynamics.

G. What are the limitations and future directions? What is left to be learned? Briefly argue the plausibility of the approach and/or what you think is essential that may have been left out.

One limitation is that the current model is simplified (two excitatory and one inhibitory units) and rate-based rather than spiking, which may limit biological realism. We should also consider long-term plasticity and neuromodulation which influence working memory as well.

Future directions should look to validate this model on real experimental data for working memory, to explore if the model accurately reflects a system akin to working memory dynamics.

In addition, we could expand our work by incorporating more complex networks, such as larger recurrent circuits or layered structures, to test whether the stability-flexibility trade-off holds in high-dimensional systems. It would also be valuable to introduce noise and spiking dynamics to examine the robustness of attractor states under natural conditions.